Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz 03 **Summer Semester 2024-2025**

DURATION: 25 MIN **FULL MARKS: 15**

CSE 4803: Graph Theory

Figures in the right margin indicate full marks of questions with their corresponding COs and POs in parentheses.

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1. Describe the necessary and sufficient condition(s) for detecting planarity of a given graph. Suppose an arbitrary graph has 6 vertices and 11 edges. Justify whether it is possible to determine whether this graph is planar. **4+4 (CO2) (PO2)**

2. Find the Independence Number, $\beta(G)$, of the following graph, illustrating the process behind how you found it: **7 (CO1) (PO1)**